Problem · Unit 5 · 10 points · 60 min · High difficulty / trap

P15 — Relative orbital motion near a target on a circular orbit: Clohessy–Wiltshire equations, drift and a counter-intuitive rendezvous

*A target T moves on a circular orbit of radius R_T around the Earth, whose centre O is the origin of the inertial frame S_0 and whose gravitational parameter is μ (the Earth's gravity on a mass m at $\mathbf{r}$ is $-m\mu\,\mathbf{r}/|\mathbf{r}|^3$). The target's angular velocity is constant, $n\,\mathbf{k}$, with the **mean motion** $n=\sqrt{\mu/R_T^3}$ ($\mathbf{k}$ normal to the orbital plane). A chaser P of mass m, subject only to the Earth's gravity, moves in the same plane close to T.*

Use the frame $S_1=\{T;\mathbf{e}_x,\mathbf{e}_y,\mathbf{k}\}$ attached to the target: $\mathbf{e}_x$ radial, pointing away from O; $\mathbf{e}_y=\mathbf{k}\times\mathbf{e}_x$ along the target's velocity (along-track); S_1 rotates with $\boldsymbol{\omega}_{10}=n\,\mathbf{k}$. The relative position of the chaser is $\boldsymbol{\rho}=x\,\mathbf{e}_x+y\,\mathbf{e}_y$ with $|\boldsymbol{\rho}|\ll R_T$. Initial conditions at $t=0$: x_0, y_0, $\dot{x}_0$, $\dot{y}_0$ (relative to S_1).

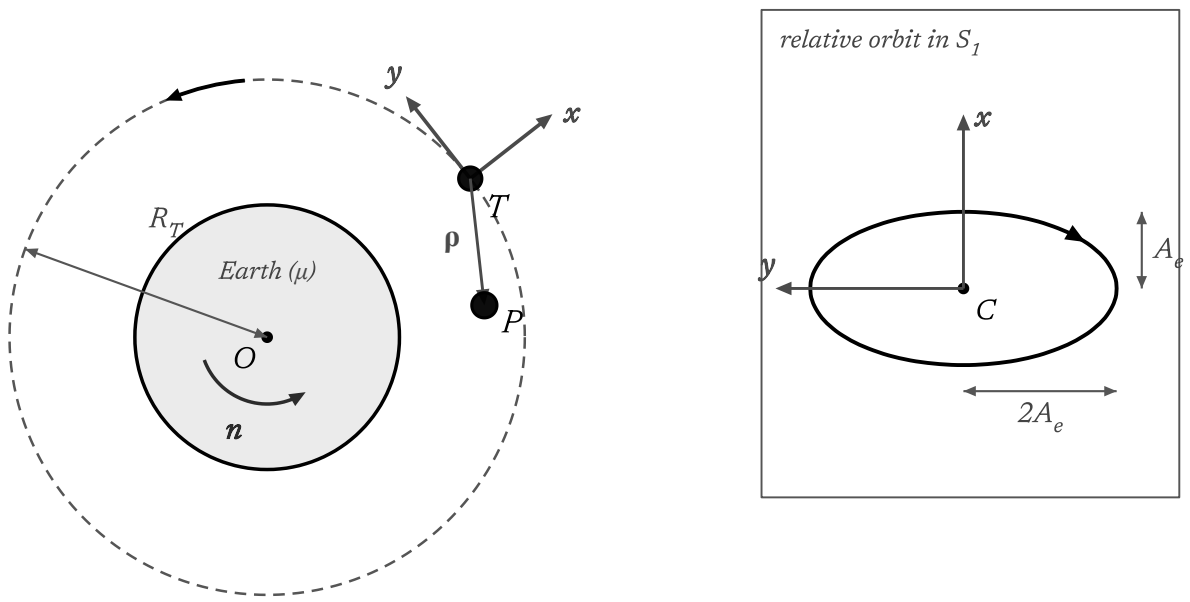


Figure 1. Target T on a circular orbit of radius R_T with mean motion n; rotating frame $S_1=\{T;\mathbf{e}_x,\mathbf{e}_y\}$ (x radial outwards, y along-track); chaser P at $\boldsymbol{\rho}=x\,\mathbf{e}_x+y\,\mathbf{e}_y$ (distance exaggerated). Inset: bounded relative orbit ($\dot{y}_0=-2nx_0$), a $2{:}1$ ellipse about its centre C, run in the sense of the arrow.

Setup

- Data: μ, R_T, $n = \sqrt{\mu/R_T^3}$, the initial conditions. Numerical part (f): $\mu = 3.986 \times 10^{14}$ m^3/s^2, $R_T = 6778$ km (low Earth orbit), chaser 1 km behind the target ($x_0 = 0$, $y_0 = -1000$ m).
- Hypotheses: point masses, planar motion in the orbital plane, S_0 inertial, no drag, no other bodies; the target's orbit is not affected by the chaser; terms of order $|\boldsymbol{\rho}|^2/R_T^2$ are neglected in the gravity difference.
- The frame S_1 has a uniformly rotating basis ($\dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$) and an accelerated origin ($\mathbf{a}_T = -n^2 R_T\,\mathbf{e}_x$).

Part (a) · 1.5 points

Degrees of freedom, frame and relative kinematics

How many degrees of freedom has P in this planar problem? Write the absolute acceleration of P in the basis of S_1 as the sum of the acceleration of T, the relative acceleration, the Coriolis and the centripetal terms, and list the inertia forces that act on P in S_1.

Step 1. Count

P is free in the orbital plane: 2 degrees of freedom, the coordinates (x, y) in S_1. There are no constraints and no reactions; the only force is gravity.

Step 2. Motion of the frame

T moves on a circle of radius R_T with constant angular velocity $n\mathbf{k}$, so its acceleration is purely centripetal: $\mathbf{a}_T = -n^2 R_T\,\mathbf{e}_x$. The basis of S_1 turns with $\boldsymbol{\omega}_{10} = n\mathbf{k}$, and $\dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$.

Change of basis: take the inertial basis $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ of S_0 with $\mathbf{i}$ along $\overrightarrow{OT}$ at $t = 0$. Then $\mathbf{e}_x = \cos nt\,\mathbf{i} + \sin nt\,\mathbf{j}$, $\mathbf{e}_y = \mathbf{k} \times \mathbf{e}_x = -\sin nt\,\mathbf{i} + \cos nt\,\mathbf{j}$, and

$$\begin{pmatrix} \mathbf{e}_x \\ \mathbf{e}_y \\ \mathbf{k} \end{pmatrix} = [R] \begin{pmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{pmatrix}, \qquad [R] = \begin{pmatrix} \cos nt & \sin nt & 0 \\ -\sin nt & \cos nt & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The rows have length $\cos^2 nt + \sin^2 nt = 1$, rows 1 and 2 are orthogonal ($-\cos nt \sin nt + \sin nt \cos nt = 0$) and row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2 nt + \sin^2 nt = +1$. Differentiating, $\dot{\mathbf{e}}_x = n\,\mathbf{e}_y = n\mathbf{k} \times \mathbf{e}_x$ and $\dot{\mathbf{e}}_y = -n\,\mathbf{e}_x = n\mathbf{k} \times \mathbf{e}_y$ (Poisson), which confirms $\boldsymbol{\omega}_{10} = n\,\mathbf{k}$.

Step 3. Relative kinematics

In S_1: $\mathbf{v}_{rel} = \dot{x}\,\mathbf{e}_x + \dot{y}\,\mathbf{e}_y$, $\mathbf{a}_{rel} = \ddot{x}\,\mathbf{e}_x + \ddot{y}\,\mathbf{e}_y$. With $\mathbf{k} \times \mathbf{e}_x = \mathbf{e}_y$, $\mathbf{k} \times \mathbf{e}_y = -\mathbf{e}_x$:

$$2\boldsymbol{\omega} \times \mathbf{v}_{rel} = 2n\big(\dot{x}\,\mathbf{e}_y - \dot{y}\,\mathbf{e}_x\big), \qquad \boldsymbol{\omega} \times \boldsymbol{\rho} = n\big(x\,\mathbf{e}_y - y\,\mathbf{e}_x\big), \qquad \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho}) = -n^2\big(x\,\mathbf{e}_x + y\,\mathbf{e}_y\big)$$

The composition of accelerations (Notes §3.2), with the Euler term zero, gives

$$\mathbf{a}_P = \big(-n^2 R_T + \ddot{x} - 2n\dot{y} - n^2 x \big)\,\mathbf{e}_x + \big(\ddot{y} + 2n\dot{x} - n^2 y \big)\,\mathbf{e}_y$$

Step 4. Inertia forces

Written as $m\mathbf{a}_{rel} = \mathbf{F} + \mathbf{F}_{in}$ (Notes §4.9), the inertia force is $\mathbf{F}_{in} = -m\mathbf{a}_T - 2m\boldsymbol{\omega} \times \mathbf{v}_{rel} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho})$: the dragging force of the origin $mn^2 R_T\,\mathbf{e}_x$, the centrifugal force $mn^2\boldsymbol{\rho}$ (about T) and the Coriolis force $-2mn\,\mathbf{k} \times \mathbf{v}_{rel}$. The Coriolis force is perpendicular to $\mathbf{v}_{rel}$ and does no work in S_1.

KEY RESULTS

- 2 degrees of freedom, (x, y); $\boldsymbol{\omega}_{10} = n\mathbf{k}$ constant, $\mathbf{a}_T = -n^2 R_T\,\mathbf{e}_x$.
- $\mathbf{a}_P = \big(-n^2 R_T + \ddot{x} - 2n\dot{y} - n^2 x \big)\mathbf{e}_x + \big(\ddot{y} + 2n\dot{x} - n^2 y \big)\mathbf{e}_y$.
- Inertia forces in S_1: $-m\mathbf{a}_T = mn^2 R_T\,\mathbf{e}_x$, centrifugal $mn^2\boldsymbol{\rho}$, Coriolis $-2mn\mathbf{k} \times \mathbf{v}_{rel}$; no Euler force.

Part (b) · 2 points

Linearised gravity and the equations of motion

Expand the gravity on P to first order in $|\boldsymbol{\rho}|/R_T$ and obtain the two equations of relative motion. Explain why the terms in R_T disappear.

Step 1. The exact gravity

$\mathbf{r}_P = \overrightarrow{OT} + \boldsymbol{\rho} = (R_T + x)\,\mathbf{e}_x + y\,\mathbf{e}_y$ and $|\mathbf{r}_P|^2 = R_T^2\big(1 + \frac{2x}{R_T} + \frac{x^2+y^2}{R_T^2}\big)$.

Step 2. First-order expansion

Keeping only first-order terms, $u = 2x/R_T$ and $(1+u)^{-3/2} = 1 - \frac{3}{2}u + O(u^2)$:

$$\frac{1}{|\mathbf{r}_P|^3} \approx \frac{1}{R_T^3}\Big(1 - \frac{3x}{R_T}\Big)$$

$$-\frac{\mu\,\mathbf{r}_P}{|\mathbf{r}_P|^3} \approx -\frac{\mu}{R_T^3}\Big(1 - \frac{3x}{R_T}\Big)\big[(R_T + x)\,\mathbf{e}_x + y\,\mathbf{e}_y\big] \approx -\frac{\mu}{R_T^2}\,\mathbf{e}_x + \frac{\mu}{R_T^3}\big(2x\,\mathbf{e}_x - y\,\mathbf{e}_y\big)$$

(the product $x \cdot 3x/R_T$ and $y \cdot 3x/R_T$ are second order and dropped). The circular orbit of T gives $n^2 R_T = \mu/R_T^2$ (Notes §8.3), i.e. $\mu/R_T^3 = n^2$.

Step 3. Newton in S_0, projected on the basis of S_1

$m\mathbf{a}_P = -m\mu\mathbf{r}_P/|\mathbf{r}_P|^3$ with $\mathbf{a}_P$ of (a):

$$\mathbf{e}_x: \quad -n^2R_T + \ddot{x} - 2n\dot{y} - n^2x = -n^2R_T + 2n^2x$$

$$\mathbf{e}_y: \quad \ddot{y} + 2n\dot{x} - n^2y = -n^2y$$

The terms $-n^2R_T$ cancel: the gravity at T is exactly what keeps T (and S_1) on its orbit, so only the **difference** of gravity between P and T (the tidal term) drives the relative motion. Likewise the centrifugal term $-n^2y$ cancels the along-track gravity component. Hence

$$\ddot{x} - 2n\dot{y} - 3n^2x = 0, \qquad \ddot{y} + 2n\dot{x} = 0$$

KEY RESULTS

- $-\mu\,\mathbf{r}_P/|\mathbf{r}_P|^3 \approx -n^2R_T\,\mathbf{e}_x + n^2\big(2x\,\mathbf{e}_x - y\,\mathbf{e}_y\big)$ (using $\mu = n^2R_T^3$).
- $\ddot{x} - 2n\dot{y} - 3n^2x = 0, \qquad \ddot{y} + 2n\dot{x} = 0.$

Part (c) · 2 points

Closed-form solution

Solve the equations of (b) for the initial conditions $x_0, y_0, \dot{x}_0, \dot{y}_0$.

Step 1. First integral

$\ddot{y} + 2n\dot{x} = \frac{d}{dt}\big(\dot{y} + 2nx\big) = 0$. Integrating from 0 to t (Barrow): $\big[\dot{y} + 2nx\big]_0^t = 0$, so

$$\dot{y} + 2nx = C, \qquad C = \dot{y}_0 + 2nx_0$$

Step 2. Radial equation

Substitute $\dot{y} = C - 2nx$ in the radial equation: $\ddot{x} - 2n(C - 2nx) - 3n^2 x = 0$, i.e.

$$\ddot{x} + n^2 x = 2nC$$

A particular solution is the constant $x_p = 2C/n$; the homogeneous solution is $A\cos nt + B\sin nt$. The initial conditions give $x(0) = 2C/n + A = x_0$ and $\dot{x}(0) = nB = \dot{x}_0$:

$$A = x_0 - \frac{2C}{n} = -3x_0 - \frac{2\dot{y}_0}{n}, \qquad B = \frac{\dot{x}_0}{n}$$

$$x(t) = 4x_0 + \frac{2\dot{y}_0}{n} - \left(3x_0 + \frac{2\dot{y}_0}{n}\right)\cos nt + \frac{\dot{x}_0}{n}\sin nt$$

Step 3. Along-track position

From Step 1, $\dot{y} = C - 2nx = -3C - 2nA\cos nt - 2nB\sin nt$. Integrating from 0 to t with the antiderivatives $\int \cos n\tau\, d\tau = \frac{\sin n\tau}{n}$ and $\int \sin n\tau\, d\tau = -\frac{\cos n\tau}{n}$ (chain rule: $\frac{d}{d\tau}\sin n\tau = n\cos n\tau$) and Barrow's rule:

$$y(t) - y_0 = -3Ct - 2nA\left[\frac{\sin n\tau}{n}\right]_0^t - 2nB\left[-\frac{\cos n\tau}{n}\right]_0^t = -3Ct - 2A\sin nt + 2B(\cos nt - 1)$$

With C, A, B:

$$y(t) = y_0 - (6nx_0 + 3\dot{y}_0)\,t + \left(6x_0 + \frac{4\dot{y}_0}{n}\right)\sin nt + \frac{2\dot{x}_0}{n}(\cos nt - 1)$$

Check: $x(0) = x_0$, $y(0) = y_0$, $\dot{x}(0) = \dot{x}_0$, $\dot{y}(0) = -6nx_0 - 3\dot{y}_0 + 6nx_0 + 4\dot{y}_0 = \dot{y}_0$.

KEY RESULTS

- First integral $\dot{y} + 2nx = C \equiv \dot{y}_0 + 2nx_0$; then $\ddot{x} + n^2 x = 2nC$.
- $x(t) = 4x_0 + \dfrac{2\dot{y}_0}{n} - \left(3x_0 + \dfrac{2\dot{y}_0}{n}\right)\cos nt + \dfrac{\dot{x}_0}{n}\sin nt$.
- $y(t) = y_0 - (6nx_0 + 3\dot{y}_0)\,t + \left(6x_0 + \dfrac{4\dot{y}_0}{n}\right)\sin nt + \dfrac{2\dot{x}_0}{n}(\cos nt - 1)$.

Part (d) · 1.5 points

Secular drift: the trap of the lower orbit

*Find the along-track drift per orbit and the condition for no drift. A chaser is placed 1 km **below** the target on its **own circular orbit** (radius $R_T - 1$ km): does it keep its position below T? Compare with a chaser released 1 km below T at rest in S_1.*

Step 1. Drift per orbit

The periodic terms of $y(t)$ repeat after one orbital period $2\pi/n$, and $x(2\pi/n) = x_0$; only the secular term changes:

$$\Delta y_{orbit} = y\Big(\frac{2\pi}{n}\Big) - y_0 = -(6nx_0 + 3\dot{y}_0)\,\frac{2\pi}{n} = -12\pi x_0 - \frac{6\pi\,\dot{y}_0}{n}$$

It vanishes iff $6nx_0 + 3\dot{y}_0 = 0$:

$$\dot{y}_0 = -2n\,x_0 \quad (C = 0)$$

Step 2. Chaser on its own circular orbit

The circular orbit of radius $R_T + x_0$ has angular rate $n' = \sqrt{\mu/(R_T + x_0)^3} = n\,(1 + x_0/R_T)^{-3/2} \approx n\big(1 - \frac{3x_0}{2R_T}\big)$. Its velocity in S_1 is the absolute velocity minus the transport velocity $n\mathbf{k} \times \mathbf{r}$: $\dot{y}_0 = (R_T + x_0)(n' - n) \approx R_T \cdot n \cdot \big(-\frac{3x_0}{2R_T}\big) = -\frac{3}{2}nx_0$ (first order). Then $C = \frac{1}{2}nx_0$, $A = -3x_0 + 3x_0 = 0$, so $x(t) = x_0$ for all t and

$$y(t) = y_0 - \tfrac{3}{2}n\,x_0\,t, \qquad \Delta y_{orbit} = -3\pi x_0$$

For $x_0 = -1000$ m: $\Delta y_{orbit} = +3000\pi = +9425$ m per orbit. **The trap:** "same circular motion, so it stays below" is wrong. The lower orbit has a shorter period (Kepler), so the chaser moves ahead of the target at $\frac{3}{2}n|x_0|$. The no-drift value $-2nx_0$ is **not** the circular-orbit value $-\frac{3}{2}nx_0$: the drift-free relative motion is the ellipse of (e), not a fixed point.

Step 3. Chaser at rest in S_1

With $\dot{y}_0 = 0$ (and $\dot{x}_0 = 0$): $\Delta y_{orbit} = -12\pi x_0 = +12\,000\pi$ m $= +37.70$ km per orbit, while x oscillates between x_0 and $7x_0$ ($x(t) = 4x_0 - 3x_0 \cos nt$, i.e. down to 7 km below T): a point "at rest" below the target is moving slower than the local circular speed and falls into an eccentric orbit.

KEY RESULTS

- $\Delta y_{orbit} = -\dfrac{2\pi}{n}\left(6nx_0 + 3\dot{y}_0\right) = -12\pi x_0 - \dfrac{6\pi\dot{y}_0}{n}$; no drift iff $\dot{y}_0 = -2nx_0$ (i.e. $C = 0$).
- Own circular orbit: $\dot{y}_0 \approx -\frac{3}{2}nx_0$, $x \equiv x_0$, $\Delta y_{orbit} = -3\pi x_0 = +9.42$ km per orbit for $x_0 = -1$ km: it **moves ahead** of T.
- At rest in S_1 ($\dot{y}_0 = 0$): $\Delta y_{orbit} = -12\pi x_0 = +37.70$ km per orbit.

Part (e) · 1.5 points

Bounded relative orbit

For $\dot{y}_0 = -2nx_0$, show that the relative trajectory is an ellipse centred on the along-track axis, with the along-track semi-axis twice the radial one. Find its centre, its period and its sense of motion.

Step 1. Substitution

With $\dot{y}_0 = -2nx_0$: $4x_0 + 2\dot{y}_0/n = 0$, $3x_0 + 2\dot{y}_0/n = -x_0$, $6nx_0 + 3\dot{y}_0 = 0$ and $6x_0 + 4\dot{y}_0/n = -2x_0$. Then (c) gives

$$x = x_0\cos nt + \frac{\dot{x}_0}{n}\sin nt, \qquad y = y_0 - \frac{2\dot{x}_0}{n} - 2x_0\sin nt + \frac{2\dot{x}_0}{n}\cos nt$$

Step 2. The ellipse

With $A_e\cos\delta = x_0$, $A_e\sin\delta = \dot{x}_0/n$ and the addition formulas, $x = A_e\cos(nt - \delta)$ and $y - y_c = -2\left(x_0\sin nt - \frac{\dot{x}_0}{n}\cos nt\right) = -2A_e\sin(nt - \delta)$. Using $\cos^2 + \sin^2 = 1$:

$$\frac{x^2}{A_e^2} + \frac{(y - y_c)^2}{4A_e^2} = 1$$

an ellipse centred at $(0, y_c)$, with semi-axes A_e (radial) and $2A_e$ (along-track), run once per orbital period $2\pi/n$.

Step 3. Sense of motion

The $\mathbf{k}$-component of $\left(x\,\mathbf{e}_x + (y-y_c)\,\mathbf{e}_y\right) \times \left(\dot{x}\,\mathbf{e}_x + \dot{y}\,\mathbf{e}_y\right)$ is $x\dot{y} - (y-y_c)\dot{x} = A_e\cos\theta\,(-2A_e n\cos\theta) - (-2A_e\sin\theta)(-A_e n\sin\theta) = -2nA_e^2 < 0$ (with $\theta = nt - \delta$): the ellipse is run in the sense of $-\mathbf{k}$. Above the target ($x > 0$) the chaser moves backwards ($\dot{y} < 0$), below it moves forwards: higher means slower.

KEY RESULTS

- $x = x_0\cos nt + \dfrac{\dot{x}_0}{n}\sin nt,\ y = y_c - 2x_0\sin nt + \dfrac{2\dot{x}_0}{n}\cos nt,\ y_c = y_0 - \dfrac{2\dot{x}_0}{n}.$
- $\dfrac{x^2}{A_e^2} + \dfrac{(y-y_c)^2}{4A_e^2} = 1$, $A_e = \sqrt{x_0^2 + \dot{x}_0^2/n^2}$; period $2\pi/n$; run in the sense of $-\mathbf{k}$ (opposite to the orbital motion).

Part (f) · 1.5 points

A low Earth orbit case: two manoeuvres

*Data: $\mu = 3.986\times10^{14}$ m^3/s^2, $R_T = 6778$ km; the chaser is 1 km behind the target, $x_0 = 0$, $y_0 = -1000$ m, at rest in S_1, and receives an impulse at $t = 0$. (i) The impulse is radial, $\dot{x}_0 = +0.5$ m/s. Describe the motion and give the position after half an orbit and after one orbit. (ii) Which **along-track** impulse $\dot{y}_0$ (with $\dot{x}_0 = 0$) brings the chaser onto the target after exactly one orbit? Which way does it point, how far does the chaser go from the orbit, and with what relative velocity does it arrive?*

Step 1. Mean motion

$$n = \sqrt{\frac{3.986\times10^{14}}{(6.778\times10^{6})^3}} = 1.1314\times10^{-3}\ \mathrm{s^{-1}}, \qquad \frac{2\pi}{n} = 5553\ \mathrm{s} = 92.6\ \mathrm{min}$$

Step 2. Radial impulse

$x_0 = 0$, $\dot{y}_0 = 0$: $C = \dot{y}_0 + 2nx_0 = 0$, so the motion is the bounded ellipse of (e), with $A_e = \dot{x}_0/n = 0.5/1.1314\times10^{-3} = 441.9$ m:

$$x = 441.9\,\sin nt\ \mathrm{m}, \qquad y = -1000 + 883.9\,(\cos nt - 1)\ \mathrm{m}$$

At half an orbit ($nt = \pi$): $x = 0$, $y = -1000 - 4\cdot441.9 = -2768$ m (the chaser rose and fell behind); centre of the ellipse $y_c = y_0 - 2\dot{x}_0/n = -1884$ m. After one orbit it is back at $(0, -1000)$ m: a radial impulse alone gains nothing.

Step 3. Along-track impulse for a one-orbit rendezvous

With $x_0 = \dot{x}_0 = 0$, (c) gives $x = \frac{2\dot{y}_0}{n}(1 - \cos nt)$ and $y = y_0 - 3\dot{y}_0 t + \frac{4\dot{y}_0}{n}\sin nt$. At $t = 2\pi/n$, $\sin 2\pi = 0$, $\cos 2\pi = 1$: $x = 0$ and

$$0 = y_0 - 3\dot{y}_0\,\frac{2\pi}{n} \implies \dot{y}_0 = \frac{y_0\, n}{6\pi} = \frac{-1000}{3 \cdot 5553} = -0.0600 \text{ m/s}$$

The trap: to catch a target 1 km **ahead**, the chaser must thrust **backwards**. The backward impulse lowers its orbit (x reaches $\frac{4\dot{y}_0}{n} = -212.2$ m at half orbit), and the lower orbit is faster, so it gains $3|\dot{y}_0| \cdot 2\pi/n = 1000$ m in one period. A forward impulse would raise the chaser and make it fall behind.

Step 4. Arrival

At $t = 2\pi/n$: $\dot{x} = 2\dot{y}_0 \sin nt = 0$ and $\dot{y} = -3\dot{y}_0 + 4\dot{y}_0 \cos nt = \dot{y}_0$. The chaser arrives at T with $\mathbf{v}_{rel} = -0.0600\,\mathbf{e}_y$ m/s, so a second impulse of $+0.0600$ m/s along $\mathbf{e}_y$ stops it at the target (total 0.120 m/s). A numerical integration of the full nonlinear two-body equations from the same initial state misses the target by about 3 m (2.8 m) after one orbit, consistent with neglecting terms of order $|\boldsymbol{\rho}|^2/R_T$.

KEY RESULTS

- $n = 1.1314 \times 10^{-3}$ rad/s, period $2\pi/n = 5553$ s $= 92.6$ min.
- (i) $C = 0$: bounded ellipse with $A_e = \dot{x}_0/n = 441.9$ m, centre $y_c = -1884$ m; at half orbit $(x, y) = (0, -2768\text{ m})$; after one orbit back at $(0, -1000\text{ m})$: no gain.
- (ii) $\dot{y}_0 = \dfrac{y_0\, n}{6\pi} = \dfrac{y_0}{3 \cdot 2\pi/n} = -0.0600$ m/s (**backwards**); x reaches $4\dot{y}_0/n = -212.2$ m at half orbit; arrival with $\mathbf{v}_{rel} = \dot{y}_0\,\mathbf{e}_y$, so a braking impulse of $+0.0600$ m/s is needed. A nonlinear two-body integration misses the target by about 3 m (2.8 m).

Interpretation and checks

- **Dimensional check:** n is s^{-1} (μ/R^3 is $\text{m}^3\text{s}^{-2}/\text{m}^3$); every term of $\ddot{x} - 2n\dot{y} - 3n^2x$ is m s^{-2}; $\dot{y}_0/n$ is m, so $x(t)$ and $y(t)$ are m; $\Delta y_{orbit} = -12\pi x_0 - 6\pi\dot{y}_0/n$ is m.
- **Limiting cases:** $\boldsymbol{\rho} = \mathbf{0}$ with zero relative velocity stays at the origin (the target itself). $n \to 0$ (no gravity, non-rotating frame): expanding $\cos nt \approx 1 - \frac{1}{2}n^2t^2$ and $\sin nt \approx nt - \frac{1}{6}n^3t^3$ in (c) gives $x \to x_0 + \dot{x}_0 t$, $y \to y_0 + \dot{y}_0 t$, uniform straight motion, as it must.
- **Physical meaning of C:** $C = \dot{y} + 2nx$ is constant; it measures the difference of orbital period between chaser and target (to first order the chaser's semi-major axis differs from R_T by $2C/n$). Equal periods ($C = 0$) means no drift; that is why the radial impulse of (f)(i) gives a closed loop.
- **Common mistakes in the exam:** (1) forgetting the acceleration of the origin $\mathbf{a}_T$ or the centrifugal term, which leaves an R_T term that does not cancel; (2) expanding $|\mathbf{r}_P|^{-3}$ with the wrong exponent ($1 - 2x/R_T$ instead of $1 - 3x/R_T$); (3) assuming that a chaser below the target on its own circular orbit, or at rest in S_1, keeps its relative position; (4) thrusting towards the target to catch it.